HYBRID UPLINK ACCESS IN IEEE 802.11ax UNDER NON-SATURATED TRAFFIC: ANALYTICAL MODELING AND PERFORMANCE EVALUATION

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ABSTRACT

The emergence of wireless systems with numerous terminals, such as Internet of Things (IoT) deployments, Smart Home Systems, and enterprise Wi-Fi networks, requires efficient channel access mechanisms. However, existing methods in IEEE 802.11ax standard (Wi-Fi 6) face performance issues under practical non-saturated traffic conditions as a result of a trade-off between scheduled and contention-based transmissions. In this paper, we study a hybrid channel access scheme which combines scheduled and random-access transmissions into a single Trigger Frame (TF). Some Resource Units (RUs) are used for scheduled access determined by Buffer Status Reports (BSRs), while the remaining RUs are allocated for random access to accommodate new terminal connections. Realistic traffic modeling is considered using Pareto traffic arrival for non-saturated conditions. The analytical model is validated via simulation to evaluate throughput, access latency, ability to provide random access, and energy efficiency. The analysis is performed for the cases of both static and dynamic scheduling mechanisms. Results demonstrate that our proposed hybrid mechanism outperforms the random-only scheme, achieving up to 3× higher throughput than random-only access in areas of dense stations; reduces access delay by over 40% compared to static scheduling; and increases energy efficiency when using dynamic scheduling.

KEYWORDS

IEEE 802.11ax (Wi-Fi 6), OFDMA, MU-MIMO, Non-Saturated Traffic, Pareto Traffic, Channel Sounding

1. INTRODUCTION

Wireless Local Area Networks (WLANs) based on IEEE 802.11 protocols have emerged as the main solution for providing high-speed Internet access. Their low deployment cost, ease of installation, and flexibility to support a wide range of applications have been key factors that have led to the widespread adoption of Wi-Fi and a rapid increase in the number of Wi-Fi-enabled devices [1], [2], [3]. Recent reports indicate that Wi-Fi accounts for more than half of global Internet traffic. The demand for these devices continues to grow rapidly, driven by the need to support emerging applications and higher data rate requirements. Additionally, the growing number of low-latency applications, including video streaming, online gaming, and Internet of Things (IoT) services, further expose performance limitations in Wi-Fi networks, particularly regarding throughput, latency, and reliability when used in crowded environments [4], [5], [6].

The IEEE 802.11ax amendment (Wi-Fi 6) was developed to support High Efficiency (HE) WLANs in dense deployment scenarios. This amendment increases network performance by achieving up to four times higher per-station throughput compared to legacy WLANs. Two primary elements of the new Medium Access Control (MAC) standard that provide the necessary level of throughput are Orthogonal Frequency Division Multiple Access (OFDMA) and Uplink Multi-User Multiple-Input Multiple-Output (UL MU-MIMO). OFDMA divides the operating channels into multiple Resource Units (RUs) ,allowing for simultaneous connections from multiple devices using OFDMA technology whether in the downlink, uplink, or both. UL MU-MIMO is for use in the upload direction providing multiple station connections to the AP simultaneously over the same frequency resource using spatial multiplexing. Additional improvements to reduce the impact of interference and improve the overall efficiency of WLANs have also been made through other enhancements such as BSS Coloring and Target Wake Time (TWT). This paper will concentrate on evaluating the performance of the new MU-OFDMA and MU-MIMO multi-user transmission mechanisms in both uplink and downlink [4] – [8].

Despite these improvements, most existing studies on IEEE 802.11ax assume saturated traffic conditions, where all stations always have data ready to transmit. However, in practical scenarios, many applications such as IoT sensing, smart home systems, and event-driven services generate bursty and intermittent traffic, which corresponds to non-saturated conditions. To model this behavior, Pareto Type II (Lomax) arrivals are considered, as they capture the heavy-tailed and self-similar characteristics of modern IoT and multimedia applications [9], [10]. The behavior of systems under such conditions differs significantly from saturated system behavior due to contention, queuing, and throughput, and which necessitates a more realistic performance evaluation [11], [12], [13].

This paper presents an analytical model supported by simulation to evaluate IEEE 802.11ax performance with regard to non-saturated traffic. In addition, we propose and evaluate a hybrid uplink channel access mechanism that integrates both scheduled and random access. In particular, we study a hybrid uplink channel access mechanism that combines scheduled and random transmissions within a unified Trigger Frame (TF). In this scheme, a portion of the Resource Units (RUs) is allocated for scheduled access based on buffer status reports, while the remaining RUs are used for random access to accommodate new or sporadic users. To the best of our knowledge, no prior work has jointly analyzed hybrid access under non-saturated traffic while accounting for MU-MIMO channel sounding overhead. The results show that the proposed approach achieves a better balance between efficiency, fairness, and signaling overhead compared to conventional schemes, especially in dense and dynamic traffic scenarios.

1.1. Literature Review

Studies on IEEE 802.11ax technology have considered MU solutions extensively. For example, in the OFDMA access mode, there have been extensive applications of Markov chain models to analyze RA [14,15], resulting in efficiency improvements in the form of frame size control [16]. Thereafter, SA schemes have been implemented [17,18] and tested via simulation tools such as NS-3 [19]. In order to counter the shortcomings of RA and SA alone, HA methods that integrate RA and SA have been suggested, usually with a combination of random access and scheduling phases [20,21]. However, most research works make the assumption of saturated traffic conditions [22].

Most recent studies on hybrid access have analyzed non-saturated (NS) conditions and how bursty traffic interacts with dynamic transition methods to enhance throughput performance [23] – [25].

MU-MIMO was initially introduced in IEEE 802.11ac only for the downlink channels [26,27] with research concerning TXOP sharing [28] and user selection [29]. MU-MIMO was extended to include both uplink and downlink in IEEE 802.11ax [26]. There are two ways of implementing uplink MU-MIMO: synchronous and asynchronous. The synchronous mode allows synchronized high-rate transmission with high timing synchronization requirements. In asynchronous mode, there is more flexibility but the system requires advanced channel sensing. Studies on scheduling in the synchronous mode [30] and simultaneous use of MU-OFDMA/MU-MIMO [31], channel sensing in asynchronous mode [32] using Markov chain model [33] and renewal process model [34] were conducted. Asynchronous UL MU-MIMO is rarely used in IEEE 802.11ax.

1.2. Motivation and Contributions

To conclude, while prior research has provided valuable insights into MU-OFDMA and MU-MIMO in the context of IEEE 802.11ax, there are still several limiting factors present. Most analytical models assume saturated traffic and rely complex analytical formulations. In addition, the majority of research has focused primarily on the metrics of throughput, whereas other important metrics such as delay, power consumption and energy efficiency remain largely unexplored. Furthermore, very limited analysis has been carried out on the use of hybrid access for non-saturation traffic with realistic channels sounding overheads in the context of MU-MIMO.

In order to facilitate a proper comparison between the existing works and our proposed research, the important aspects and limitations of the literature that are most pertinent to the study are presented in Table 1. As can be observed from the table below, the majority of existing studies are limited to saturated traffic and throughput only. The study which is most relevant among all the studies examined is [25] which lays down the basics of hybrid access with respect to nonsaturated Poisson traffic. It does not cover Pareto traffic, channel sounding, and important performance parameters including delay, energy consumption, and energy efficiency.

Building upon the framework in [25], this paper makes the following key contributions. First, a comprehensive performance evaluation is conducted on non-saturated traffic conditions using a Pareto method of evaluation to capture the trade-offs among key performance indicators including throughput, access delay, Random Access Capability (RAC), energy consumption, and energy efficiency for the various types of scenarios that represent the system’s operation in unsaturated network conditions. A comprehensive review of the performance evaluation results performed on non-saturated traffic conditions provides a more realistic portrayal of actual network behavior, thus giving greater insight into the various mechanisms that operate within the system under normal operating conditions.

Secondly, a dynamic Resource Unit (RU) allocation scheme is proposed that the Resource Units (RUs) are allocated based on Buffer Status Reports (BSRs) rather than conventional static allocation the Resource Units (RUs) to users before receiving Buffer Status Reports (BSRs). Through the use of real-time traffic demands, the dynamic Resource Unit (RU) allocation method will enable a more effective manner of using or allocating resources based on from both the minimization of underutilization and the minimization of congestion and thus maximize the system response time and overall system performance. The results demonstrate that use of the dynamic Resource Unit (RU) allocation scheme will produce a more efficient means of resource management in comparison to the use of the traditional means of static allocation of Resource Units (RUs).

Table1. Comparison of existing studies and the proposed work.

Reference	Traffic Condition	Hybrid Access	MU-MIMO Analysis	CSI Overhead	Metrics Analysis
[14],[15],[16]	Saturated	RA Only	✗	✗	Throughput
[17]	Non Saturated	RA Only	✗	✗	Throughput Delay
[18]	Saturated	RA+SA Static Allocation	✗	✗	Throughput
[19]	Saturated	RA+SA Static Allocation	✓	✓	Throughput, Delay, Power Efficiency, RA Capability
	Non Saturated	RA Only, Dynamic Access	✗	✗	Throughput
[28]	Saturated	RA Only	✓	✗	Throughput
Our work	Non Saturated	RA+SA Static & Dynamic Allocation	✓	✓	Throughput Delay RA Capability Power Consumed Energy Efficiency

Finally, a unified MU-MIMO framework is developed that includes overhead resources needed to accurately quantify the cost of Channel State Information (CSI) acquisition in accordance with IEEE 802.11ax. The unified MU-MIMO model also takes into consideration the overhead introduced during the process of channel sounding which is required for an appropriately reliable estimate of CSI and is required for uplink MU-MIMO operation in an MU-MIMO environment. By including both transmission components as well as overhead components together as part of the same unified MU-MIMO system configuration within a single framework, a more realistic and comprehensive evaluation framework is provided to evaluate the costs required to acquire CSI for various MU-MIMO system configurations.

1.3. Organization

The remainder of this paper is structured as follows: the system model and analytical framework are defined in Section 2; the transmission and hybrid-access mechanisms are addressed in Section 3; performance measure definitions are discussed in Section 4; and result and discussion sections are provided in Section 5. Finally, the paper's conclusion is introduced in section 6.

2. SYSTEM MODEL & ANALYTICAL MODELING

2.1. Network Assumptions

Consider a Basic Service Set (BSS) IEEE 802.11ax (Wi-Fi 6) network consisting of one Access Point (AP) and multiple connected Stations (STAs). To focus exclusively on MAC-layer behavior, an ideal physical (PHY) layer is assumed, providing error-free transmission and perfect channel estimation. Across all devices, a subset is scheduled using the Scheduled Access (SA) mechanism, while the remaining STAs rely on Random Access (RA) through uplink trigger-based contention, where $n_T = n_{SA} + n_{RA}$.

The AP dynamically allocates uplink Resource Units (RUs) over the available channel bandwidth B, which is partitioned into equal-sized subchannels. Each RU employs the same

Modulation and Coding Scheme (MCS) index to ensure fairness among contending STAs.Let the total number of RUs be divided into N_{sa}scheduled RUs and N_{ra}random-access RUs such that$N_{ru} = N_{sa} + N_{ra}$.

Unlike saturated traffic models, the proposed system considers non-saturated traffic conditions, where each STA may or may not have packets available for transmission during agiven Trigger Frame (TF) cycle. Packet arrivals at the STAs are modeled as a stochastic process such that the packet inter-arrival times are distributed according to a Pareto distribution to account for the characteristics of bursty and self-similar traffic.

Each STA is equipped with a finite buffer capable of holding a maximum of P_{max} packets. A bulk-service aggregation mechanism is assumed, allowing up to P_{Agg} packets to be transmitted in a single successful attempt. The AP operates with N antennas, and each STA with P antennas, enabling MU-MIMO grouping when applicable.

The minimum contention window is denoted as W_{min}, and exponential backoff is applied for failed RA transmissions up to a retry limit m_{limit}.

2.2. Non-Saturated Traffic Condition

Contrary to normal saturated models, where there are packets in all time units in the buffer for STAs to transmit, in a practical WLAN environment, there are random arrivals of packets, making STAs switch between being idle, being in backoff state, and transmitting the packets in accordance with the buffer content. Once the service period expires, the STA enters an idle state in case the buffer is empty or goes into backoff again in case the buffer still has packets.

Let q denote the probability that a STA has at least one packet remaining after service, and r denote the probability that a packet arrives during the backoff countdown. If the buffer is empty after service, the STA waits for an average ofTF-cycles before a new packet is produced [35].

Hence, the system is renewed after an expected interval of $\frac{1-q}{r}$ time slots. Both q and r are derived as functions of the selected traffic model, according to their statistical characteristics.

2.3. Pareto Traffic Model

In contrast to the traditional memory less Poisson traffic model, we require a better traffic model to model bursty Internet traffic that occurs in the real world. The self-similarity [36] of the WLAN traffic is captured by the realistic, fat-tailed Pareto traffic framework. The Pareto type II (Lomax distribution) [37] has been taken into consideration for analytical modeling of packet inter-arrival time. The Lomax (Pareto Type II) distribution is characterized by its cumulative distribution function (CDF) $F(t)$ and probability density function (PDF) $f(t)$for $t > 0$ as follows$F(t) = 1 - \left(\frac{\beta}{\beta+t}\right)^{\alpha}, f(t) = \frac{\alpha}{\beta}\left(\frac{\beta}{\beta+t}\right)^{\alpha+1}$'where $\alpha > 0$ is the shape parameter and $\beta > 0$ is the scale parameter. The expected arrival time is:At any packet departure instant, the probability that the buffer is non-empty is approximated as:$q = \min\,(1, \lambda\, E[D])$,and the probability that a packet arrives during the countdown interval $t_s$is:$r = 1 - (\frac{\beta}{\beta+t_s})^{\alpha}$.Let S_n is the

sum of n independent and identically distributed (IID) Pareto random variables. Its PDF is given by [37]:

$$f_{S_n}(t) = \frac{1}{n\beta}\int_0^{\infty} e^{-(1+\frac{t}{n\beta})v}\psi_{m,n}(\frac{v}{n})dv \quad \text{. For v} > 0 \tag{1}$$

$$\psi_{m,n}(v) = v(-1)^{n+1}m^n \sum_{r=0}^{\lfloor (n-1)/2 \rfloor} (-\pi^2)^r \times \binom{n}{2r+1}(\frac{v^m}{m!})^{2r+1}\mathrm{Ei}_{m+1}^{n-2r-1}(v) \tag{2}$$

$$\mathrm{Ei}_m(y) = \frac{y^{m-1}}{(m-1)!}(Eu + \ln y - \sum_{r=1}^{m-1}\frac{1}{r}) + \sum_{\substack{r=0 \\ r\neq m-1}}^{\infty}\frac{y^r}{(r-m+1)r!} \tag{3}$$

Where $E_u = 0.5772156649$ is Euler's constant. The probability that n packets have been received by time t is calculated [38]:

$$P(N(t) = n) = \frac{f_{S_{n+1}}(t)}{\lambda} \tag{4}$$

Accordingly, the average number of packets generated during the average service time $t = E[D]$is:

$$n_f(t) = \sum_{n=0}^{\infty} n\, P(N(t) = n) \tag{5}$$

This formulation effectively models the variability and burstiness of non-saturated WLAN traffic, providing a realistic analytical foundation for IEEE 802.11ax performance evaluation.

2.4. Actual Packet Transmission and Aggregation

IEEE 802.11ax employs a multiple-TID-based frame bundling approach, allowing STAs to transmit multiple packets associated with distinct Traffic Identifiers (TIDs) within a single PHY overhead interval [20].The transfer data rate of a device is given by:

$$D_r = (N_{SS})(B_{mcs})(C_{mcs})(N_{sc}^{RU}) \text{ (bits/symbol)} \tag{6}$$

Where N_{SS} is the number of spatial streams, $B_{mcs}$is the number of bits per OFDM symbol, C_{mcs} is the coding rate, and N_{sc}^{RU} is the number of data subcarriers per RU [39]. Thus, the effective data rate (in bps) is: $R_{eff} = \frac{D_r}{T_{sym}}$, where $T_{sym}$is the OFDM symbol duration. If L_p is the packet payload (bits), with MAC header $L_h$and tail bits L_t, the maximum number of bundled packets in a TXOP of duration $t_d$is:

$$P_{Agg} = \min\ (P_{max}, \lfloor \frac{D_r(T_d - T_{PHY})/T_{sym}}{L_t + L_h + L_p} \rfloor) \tag{7}$$

Where P_{max} is the maximum aggregation limit in IEEE 802.11ax [20]. Under non-saturated traffic, the number of packets sent successfully in a service period is [25]:

$$N_{Tx} = \min\ (n_f, P_{Agg}) \tag{8}$$

Where $n_f$is the number of packets generated during $E[D]$.This model follows a bulk-service queuing process [38], forming the foundation for the dynamic scheduling framework discussed later.

2.5. Transmission Probability Derivation

This section presents the analytical derivation of the transmission probability (τ) and conditional collision probability (γ) for a device operating under IEEE 802.11ax uplink hybrid access. The model is developed using the FPA Analysis method [39], which provides a unified and computationally efficient alternative to conventional Markov chain modeling [40], [41]. The transmission probability τ denotes the probability that a device attempts transmission over the channel, while γ represents the conditional probability that a transmission attempt results in a collision. Let $b_i$denote the average number of backoff (BO) slots experienced by a STA at contention stage i, $B^{(j)}$denote the total backoff duration experienced by the jth BSR packet, and $T^{(j)}$denote the number of transmission attempts required for the j^{th}packet to be to either be successfully delivered or discarded. Since each packet transmission process forms a renewal process, the system can be analyzed using the Renewal Reward Theorem [38], where $T^{(j)}$represents the payoff corresponding to a renewal interval of duration $B^{(j)}$.The expected number of sending trials necessary for a packet to be successfully transmitted or discarded is expressed as $E[T] = \sum_{i=0}^{m+f} \gamma^i$ Similarly, the average duration required for these packets to complete either successful or failed transmissions is:$E[B] = \sum_{i=0}^{m+f} b_i\gamma^i$.According to the Renewal Reward Theorem, the transmission probability for non-saturated traffic is modified as:

$$\tau = \frac{\sum_{i=0}^{m+f} \gamma^i}{\sum_{i=0}^{m+f} b_i\gamma^i + \frac{1-q}{r}} \tag{9}$$

For IEEE 802.11ax, the non-uniform backoff decrement across Random Access (RA) resource units modifies $b_i$compared to legacy systems. If a STA selects an initial backoff counter of 0, it requires one BO slot to transmit. For all other randomly generated counter $i \geq 0$, the required BO slots are approximately $\left\lceil \frac{i}{N_{ra}} \right\rceil$. The average backoff duration at stage k is therefore:

$$b_k = \frac{1}{W_k}\left[1 + \sum_{i=0}^{W_k-1}\left(\frac{i}{N_{ra}} + \Delta_i^k\right)\right] \tag{10}$$

where $\Delta_{i,k} = \lceil i/N_{ra} \rceil - \{i/N_{ra}\}$, for $1 \leq i \leq W_k - 1$.Simplifying, the $E[B]$ term using b_k,we get:

Table 2. Maximum number of RUs for different channel widths.

RU size	Channel Bandwidth			
	20 MHz	40 MHz	80 MHz	160 MHz
26 tone	9	18	37	74
52 tone	4	8	16	32
106 tone	2	4	8	16
242 tone	1	2	4	8
484 tone	-	1	2	4
996 tone	-	-	1	2

$$\sum_{k=0}^{m+f} b_k\gamma^k = \sum_{k=0}^{m+f}\left(\frac{\gamma^k}{W_k} + \frac{W_k-1}{2N_{ra}}\gamma^k + \frac{N_{ra}-1}{2N_{ra}}\gamma^k\right) \tag{11}$$

Substituting this expression into the transmission probability equation yields a non-linearmapping $\gamma = F(\gamma)$.According to Brouwer's Fixed-Point Theorem [42], such a system admits a unique fixed-point solution for$\gamma \in [0,1]$, provided that $b_k$is a non-decreasing sequence.The steady-state transmission probability τ is then obtained by substituting the fixed-point value of γ into the above relation.

3. MU-TRANSMISSION & HA MECHANISM

3.1. Overview of OFDMA

OFDMA has been introduced in IEEE 802.11ax, which revolutionizes the mechanism of channel access in IEEE 802.11ax. In previous versions of the standard, where the channel was divided into only one station, OFDMA breaks the channel into many Resource Units (RUs). RU has a certain number of subcarriers or tones. One important physical layer improvement in IEEE 802.11ax is the decrease in subcarrier spacing by a factor of four along with an increase in the duration of OFDM symbols by four times, leading to more subcarriers in the same bandwidth.

The standard sizes of Resource Units in IEEE 802.11ax are 26, 52, 106, 242, 484, 996, and 2×996 tones. The number of RUs available depends on the bandwidth of the channel as well as the selected RU size (see Table 2). It is also important to note that multiple RUs can be with different sizes utilized in one OFDMA transmission.

3.2. Uplink OFDMA Frame Structure and Access Mechanisms

Figure 1 demonstrates the UL Multi-User hybrid access model employed by the IEEE 802.11ax protocol family, illustrating the sequential exchange of control and data frames between the AP and the connected STAs. The figure illustrates the two-dimensional approach, wherein the x-axis shows the ordering of frame exchange, and the y-axis shows the RU allocation in the frequency domain.

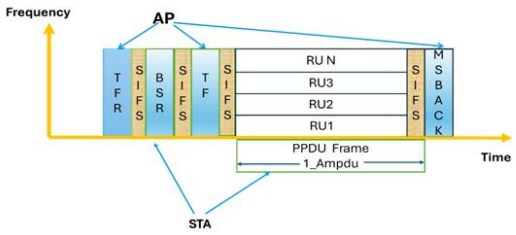


Figure 1. MU-Uplink Hybrid Access System Model.

In the initial process, the Access Point transmits the Trigger Frame Request (TFR) control frame and share information on RU allocation with STAs. After sending the TFR, any STA with pending data ready to be transmitted in the UL initiates the Distributed Coordination Function (DCF) process by lowering its backoff (BO) counters. Those STAs whose BO counter values drop to zero will begin sending Buffer Status Reports (BSRs).

The contention resolution procedure employs the following algorithm: In case of collisions during data transmission, the BO index (i) is incremented by STAs until it reaches the value of $m + f + 1$, which signifies the maximum number of retransmission attempts allowed. The contention window size is determined using a binary exponential backoff algorithm, where

$W_i = 2^i W_{min}$ for $i \leq m$, and $W_i = 2^m W_{min}$ for $m+1 \leq i \leq m+f$ [40]. In RU allocation, two types of allocation are considered:

1. Scheduled Access (SA): RUs will be allotted to stations that are allotted in advance.
2. Random Access (RA): The remaining RUs will be allotted on contention basis.

After successful BSR transmission, RA RUs can assume any of the following three conditions:

1. Success – A single STA transmission takes place on the RU.
2. Failure – Collisions occur among multiple STA transmissions on the RU.
3. Idle – There is no STA transmission activity on the RU.

The AP successful BSRs (Buffer Status Reports) are then mapped to their corresponding STAs for data transmission by scheduling this information via a subsequent Trigger Frame. Such an assignment qualifies static scheduling since RU assignments are made without changing irrespective of any changes in buffer state across TF cycles. Conversely, dynamic scheduling can also be implemented whereby the AP makes decisions about RU allocation for each individual STA using the BSR feedback. The process is summarized below:

- In the initial cycles, all RUs are used for RA to determine BSR
- As BSR information is gathered, the AP decides on STAs that will require scheduled resources
- The value of Slots ($Slots_{Sch}$) will be a function of packets ($Rem_{pkts} = n_f - \min(n_f, P_{Agg})$) [25].
- In the case of scheduled cycles, STAs are not allowed in RA.

This strategy greatly lowers the data access latency for data-intensive STAs, at the same time keeping open opportunities for new transmitters due to the continuous presence of RA-RUs. With resource allocation being dynamically adjusted based on buffer status, the scheduler effectively assigns resources to multiple TF-Cycles, thus ensuring maximum throughput and fair resource distribution between stations. The scheduled STAs send uplink traffic using frame aggregation, and then the AP sends MS-BACK control messages to acknowledge successful receptions of all packets.

The complete cycle of frame exchanges can be referred to as one TF-Cycle, which is the basic unit of time in terms of the operation of the system. As depicted in Figure 2, the time required for one TF-Cycle can be expressed using the following equation:

$$t_s = T_{TFR} + T_{TF} + T_{BSR} + T_{PPDU} + T_{MB} + 4T_{SIFS} \quad (12)$$

Where the parameters a, b, c, d, and e denote the time required for TFR, TF, BSR, PPDU.

3.3. MU-MIMO Integration and Channel Sounding

The implementation of MU-MIMO is one of the great developments of IEEE 802.11ax that uses spatial diversity along with frequency and time domain utilized by OFDMA. As compared to the single-user MIMO that provides service for one station for each transmission opportunity, MU-MIMO supports simultaneous communication with several stations through spatial streams. For our study, we use a combination approach in which MU-OFDMA operates during SA. Let N denote the number of antennas at the AP; to fully utilize the MU-MIMO capability, all antennas are assumed active during scheduled transmissions. Consequently, the number of

simultaneously served STAs can be expressed as $n_{sa} = N_{sa} \times N$, where N_{sa} represents the number of resource units (RUs) allocated for scheduled access.

For efficient MU-MIMO operation, the AP requires accurate CSI to separate users spatially and minimize inter-user interference. Hence, a channel sounding procedure is periodically performed, initiated by the AP through a Null Data Packet Announcement (NDPA) frame followed by a Null Data Packet (NDP) frame, which allows STAs to estimate the channel. Subsequently, the AP triggers a subset of STAs to transmit their CSI feedback using the MU-OFDMA capability, enabling up to U STAs to report their CSI simultaneously depending on the number of available RUs. When the total number of STAs exceeds U, multiple sounding rounds (denoted by K) are required, each consisting of a Trigger frame followed by uplink CSI PPDUs. The total duration of the sounding process is given by [25]:

$$T_{csi} = T_{NDPA} + T_{NDP} + 2T_{SIFS} + K(T_{Trigger} + 2T_{SIFS} + T_{PPDU}^{csi}) \quad (13)$$

where T_{NDPA}, T_{NDP}, $T_{Trigger}$, and T_{PPDU}^{csi} denote the durations of the respective control frames. By correctly implementing CSI acquisition using this method, the AP can maintain proper spacing between users, optimize throughput, and reduce collision chances during uplink transmission in MU-MIMO. If the channel-sounding process occurs at periodic intervals of λ_{csi}, the total time available for sending actual uplink traffic in any period is . Hence, the normalized overhead caused by the channel-sounding process is determined as:

$$O_{vf} = 1 - \lambda_{csi} T_{csi} \quad (14)$$

4. KEY PERFORMANCE METRICS ANALYSIS

4.1. Throughput

The throughput metric is considered to be the most important performance parameter in the IEEE 802.11ax technology. It measures the total amount of payload bits transferred successfully within one unit time for SA and RA stations.

The mean number of successful transmissions per TF-cycle is given by:

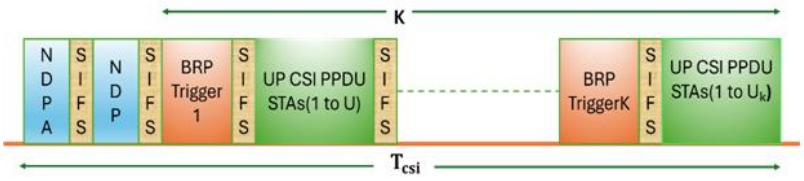


Figure 2. Channel Sounding Operation.

$$T_{MU}^{\text{Total}} = T_{RA} + n_{SA} \text{ where and } T_{RA} = N_{ra}\, n_{RA} \left(\frac{\tau}{N_{ra}}\right)\left(1 - \frac{\tau}{N_{ra}}\right)^{n_{RA}-1} \quad (15)$$

Accordingly, the total throughput of the MU-OFDMA based Hybrid Access (HA) mechanism can be expressed as [25]:

$$S_{MU} = N_{Tx} T_{MU}^{\text{Total}} \frac{L_p}{t_s} \quad (16)$$

In MU-MIMO operation, $n_{SA} = N_{SA}\ N$ .Under channel sounding overhead is considered, the achievable uplink throughput is reduced and can be expressed as:

$$S_{ul}^{\mathrm{CSI}} = O_{vf} \cdot S_{MU} \tag{17}$$

4.2. RA Capability

RA Capability (RAC) measures how quickly new stations can join the network through random access. It represents the percentage of successful random-access attempts during one transmission TF cycleis calculated as:

$$RAC_{MU} = \frac{T_{RA}}{n_{RA}} \tag{18}$$

4.3. Uplink Delay

Uplink delay refers to the duration from when a Random Access (RA) station begins contention until it successfully transmits its packet [42]. Let X_i represent the number of back-off (BO) slots experienced by a packet in the ith BO stage. The station selects its BO counter value from a uniform distribution $U(0, cw-1)$, where $cw = 2^i W_{\min}$ for $i \le m$, and $cw = 2^m W_{\min}$ for $i > m$. This counter decreases by R_{ra} in each subsequent time slot until it reaches zero.

The number of BO slots is n if the station selects a BO value within $(nN_{ra}, (n+1)N_{ra}]$, where $2 \le n < Q$, and $Q = \lceil (cw-1)/N_{ra} \rceil$ represents the upper bound on the number of back-off slots incurred by a device when it randomly selects a counter value between $((Q-1)N_{ra}, cw-1]$. The probability mass function (PMF) of X_i is given by:

$$P(X_i = 1) = \frac{N_{ra}+1}{cw}, P(X_i = n) = \frac{N_{ra}}{cw} \text{ for } n \in [2, Q), P(X_i = Q) = \frac{(cw-1)-(Q-1)N_{ra}}{cw} \tag{19}$$

The expected number of BO slots in stage i is: $E[X_i] = \frac{1}{cw}(1 + Q \cdot r + N_{ra} \sum_{e=1}^{Q-1} e)$ where $r = (cw-1) - N_{ra}(Q-1)$ denotes the remainder of the division.

The mean uplink delay is:

$$[D] = t_s \sum_{c=0}^{m+f} \Big(\sum_{i=0}^{c} E[X_i] \Big) p_c \tag{20}$$

where $p_c = \frac{(1-\gamma)\gamma^c}{1-\gamma^{m+f+1}}$ is the probability that the packet experiences c collisions and succeeds in the $(c+1)^{th}$ time slot, given it is not discarded.

Under channel sounding, STAs experience additional delays. During $U[D]$, approximately N_{csi} channel sounding operations may be performed. The value of N_{csi} can be determined by satisfying the condition:

$$P(N_{csi}-1)\frac{1}{\lambda_{csi}} < N_{csi}T_{csi} + E[D] < \frac{N_{csi}}{\lambda_{csi}} \tag{21}$$

Thus, the total delay , including the sounding process, is [19]:

$$PE[D]_{csi} = N_{csi}T_{csi} + E[D] \tag{22}$$

4.4. Energy Efficiency & power consumption

Energy efficiency measures the network's ability to deliver data with minimal energy consumption, defined as the number of successfully transmitted bits per unit energy.

Let n_{ATT} and n_{SUC} be the number of RA STAs that attempt and succeed in transferring their Buffer Status Report (BSR) during one TF cycle, respectively: (23)

$$n_{ATT} = n_{RA}\tau, n_{SUC} = \tau n_{RA}(1 - \frac{\tau}{N_{ra}})$$

In various stages of the TF process, operational states of devices differ. While exchanging TF-R and TF Control frames, the active RA devices operate in receiving mode. Those devices whose backoff counters reach zero send their BSR frames, while other active devices remain in the idle state in the idle state. For transmitting aggregated uplink frames, only successful Devices will be transmitted and then switch to receiving mode. During intermediate SIFS intervals, devices will continue in idle state.

Let p_r, p_t, p_i, p_d represent the average duration a device in reception mode, transmit mode, idle mode, and low-power mode, respectively, during one TF cycle [25]:

$$p_d = \frac{n_T - n_A}{n_T} t_s + \frac{d_{tf}}{t_f} \quad (24)$$

$$p_i = \frac{n_{RA} - n_{SUC}}{n_T}(T_{PPDU} + T_{MB}) + \frac{n_{RA}}{n_T}(4T_{SIFS}) + \frac{n_{RA} - n_{ATT}}{n_T}(T_{BSR} + T_{TFR}) \quad (25)$$

$$p_r = \frac{n_{RA}}{n_T} T_{TFR} + \frac{n_{ATT}}{n_T} T_{TF} + \frac{n_{SUC}}{n_T} T_{MB} \quad (26)$$

Let P_r, P_t, P_i, P_d denote the power consumed by a device in reception mode, transmit mode, idle mode, and low-power mode, respectively. The average energy consumed per RA and SA STA during a single TF cycle is:

$$E_{RA} = p_r P_r + p_t P_t + p_i P_i + p_d P_d \quad (24)$$

$$E_{SA} = \frac{n_{SA}}{n_T}[P_r(T_{TFR} + T_{TF} + T_{MB}) + P_i(4T_{SIFS} + T_{BSR}) + P_t T_{PPDU}] \quad (25)$$

The total power consumption per device is

$$P_t = \frac{E_{SA} + E_{RA}}{t_s} \quad (26)$$

Let S_{per} be the per-device throughput for a given TF cycle, is given by:

$$S_{per} = \frac{S_{ul}}{n_T} \quad (27)$$

The energy performance of the MU-OFDMA system is:

Table 3. System Model Parameters and Analytical Variables.

Symbol	Description	Value
N_{ra}	Total number of Rus	16
W_{min}	Minimum Contention Window Size	64
m	Maximum Backoff Stage	5
m_{limit}	Maximum Retry Limit	6
λ	Packet Arrival Rate	100
M	Number of AP Antennas	8
B	Channel BW (MHz)	160
MCS	Modulation and Coding Scheme Index	11
B_{mcs}	Bits per symbol for a given modulation	10
C_{mcs}	Coding Rate Corresponding to MCS	5/6
L_p	Payload Size (bits)	12000
L_h	MAC header (bits)	320
L_t	MPDU Delimiter length in bits	32
P_{max}	Maximum A-MPDU Aggregation Size (packets)	256
T_{sym}	OFDM Symbol Duration (µs)	16
T_{PHY}	Physical Layer MU Header Duration (µs)	228
T_{TFR}	Trigger Frame_Request Duration (µs)	70
T_{TF}	Trigger Frame Duration (µs)	100
T_d	Maximum Duration of PPDU (ms)	5.4
L_{MB}	Multi-STA Block ACK Length (bits)	40
T_{PPDU}^{CSI}	UL CSI PPDU Duration (ms)	4.2
T_{NDP}	Null Data Packet Duration(µs)	168
T_{NDPA}	Null Data Packet Announcement Duration (µs)	449
$T_{Trigger}$	Beamforming Report Poll Trigger (µs)	650
f_c	Carrier Frequency (GHz)	2.5
w_{sc}	Subcarrier Spacing (kHz)	78.125
T_{SIFS}	Short Inter-Frame Space (µs)	16
T_{BSR}	Buffer Status Report frame duration(µs)	25
P_d, P_i	Power consumed during Doze and Idle modes	150,300
P_t, P_r	Power consumed during transmission and reception (mW)	1500,600

$$\eta = \frac{S_{per}}{P_t} \tag{28}$$

Under channel sounding, the energy consumption model is extended as follows. As in [25],E_{CSI} be the amount of energy required by each device for performing a single channel sounding process. In the initial frames of the NDPA, NDP, and Trigger frames, the SA devices remain in receive mode, whereas the SA devices go into transmit mode when UL CSI PPDU is transmitted. Energy required for a single channel sounding:

$$\begin{aligned} E_{CSI} = \frac{1}{n_T}\{&P_r(T_{NDPA} + T_{NDP} + KT_{Trigger})n_{SA} + P_d(T_{csi})(n_T - n_{RA} - n_{SA}) \\ &+P_l(1+2K)T_{SIFS}\ (n_{RA} + n_{SA}) + P_l(T_{NDPA} + T_{NDP} + KT_{Trigger})n_{RA} \\ &+K[P_t T_{PPDU}^{CSI}\ U + P_l(n_{ra} + n_{sa} - U)T_{PPDU}^{CSI}]\} \end{aligned} \tag{29}$$

Consequently, the overall power used per STA during the channel sounding process for a specific TF Cycle is:

$$P_{Total}^{CSI} = \lambda_{csi} E_{CSI} + (1 - \lambda_{csi} T_{csi}) \frac{E_{RA} + E_{SA}}{t_s} \tag{30}$$

Thus, the corresponding energy efficiency is:

$$\eta_{CSI} = \frac{S_{per}^{CSI}}{P_{Total}^{CSI}} \tag{31}$$

5. RESULTS AND DISCUSSION

In this section, extensive MATLAB simulations are conducted to verify the MU-based HA system model. In each iteration, which is 20 in our experiment, the number of TF-Cycles will be taken to be 10000. The parameters and their default values for 802.11ax simulation are listed in Table 3. From the analysis, the theoretical results are close to the simulation results

5.1. Impact of Packet Arrival Rate (λ) on Throughput and Efficiency

Figure 3 below shows the effect of an increasing packet arrival rate (λ) on the throughput of the system under the Pareto traffic scheme. With a rising λ, more packets will accumulate in the buffer of the STAs, resulting in N_{tx} packets that can be transmitted per TF-cycle. Thus, the performance and efficiency of resource usage of the system will be greatly enhanced.

5.2. Impact of Resource Partitioning Between Random and Scheduled Access

Figure 4 shows the effect of changing the random access (RA) resource ratio on the performance of the entire system based on five metrics, namely throughput, probability of success for random access, delay, energy consumption, and energy efficiency. Given that the total number of RUs remains constant, increasing the RA resources automatically decreases the scheduled access (SA) resources. This implies that there is an inherent trade-off between the number of RA resources and SA resources. From the results, it is clear that as RA resources increase, it is easier to achieve successful access when the network load is low. But, as the number of stations increases, more collisions/contention events will arise, leading to lower throughput, delays, and energy efficiency. On the other hand, increasing the SA resources increases throughput and energy efficiency but at the cost of longer delays and less random-access success.

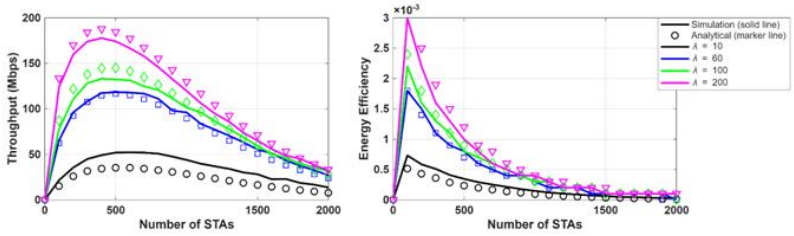


Figure 3. Impact of Pareto arrival Rate on Throughputand Energy Efficiency (in Mbps/mw): $R_{sa} = 0$.

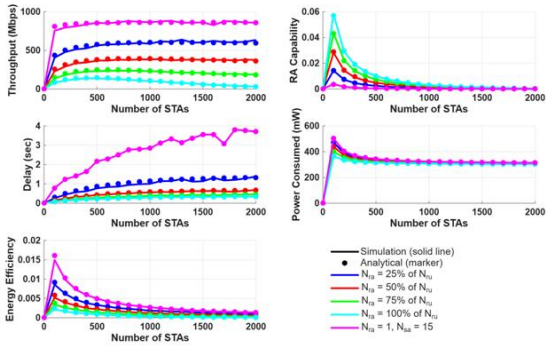


Figure 4.Impact of the distribution in Nra and Nsa for an MU-OFDMA based HA mechanism: Nru = 16. Throughput, RA capability, Delay, Power consumed, Energy efficiency (in Mbps/mw).

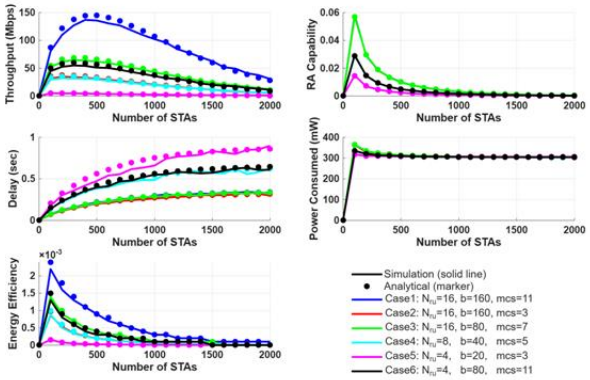


Figure 5.Joint Impact of Physical Layer Configuration and Resource Allocation. Throughput, RA capability,.Delay, Power consumed, Energy efficiency (in Mbps/mw).

5.3. Impact of PHY and Resource Configuration on System Performance

Fig. 5. illustrates the impact of physical layer and resource configuration on system performance under non-saturated Pareto traffic. Six configurations are analyzed by jointly varying channel bandwidth, number of resource units (Nru), and modulation and coding scheme (MCS).Configurations with larger bandwidth and higher MCS consistently achieve higher throughput, with Case 1 (Nru = 16, BW = 160 MHz, MCS = 11) representing the performance upper bound. Reducing MCS (Case 2) or channel bandwidth (Case 3) significantly lowers throughput, emphasizing the importance of spectral efficiency and sufficient bandwidth. More constrained configurations with fewer resource units and narrower bandwidth (Cases 4 and 5) exhibit the lowest throughput and rapidly increasing delays as station density rises. Case 6

demonstrates that employing a high MCS can partially compensate for limited bandwidth and resources, resulting in improved throughput and reduced delay. Overall, these results confirm that system performance under bursty non-saturated traffic is determined by the combined configuration of bandwidth, MCS, and resource allocation, highlighting the importance of cross-layer PHY–MAC design in IEEE 802.11ax networks.

5.4. Impact of the Number of AP Antennas on System Performance

Figure 6. examines the effects of different AP antenna numbers on system performance when there is non-saturated traffic. A higher number of antennas positively influences system throughput because of increased spatial multiplexing and better simultaneous data transmission. The difference between delay values for various numbers of antennas is low since medium access still affects delay behavior. Therefore, the delay curves partly overlap each other in the case of high numbers of stations. Even though power consumption rises as the number of antennas increases, the growth in throughput is faster than that of power consumption.

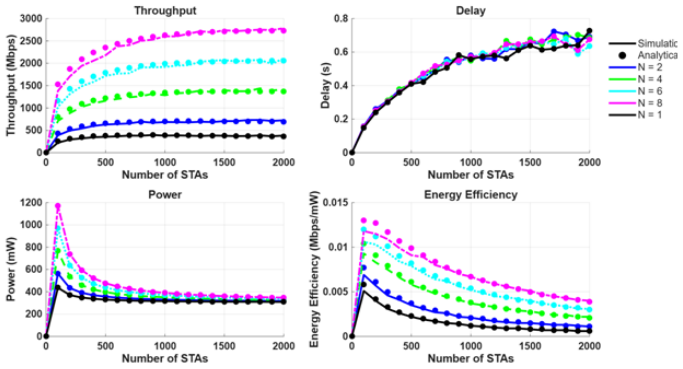


Figure 6.Effect of the Number of AP Antennas on Throughput, Delay, Power, and Energy Efficiency (in Mbps/mw). The system operates in OFDMA mode for a single antenna, while enabling MU-MIMO when multiple antennas are used :Nsa=8,Nra=8,lcsi=5

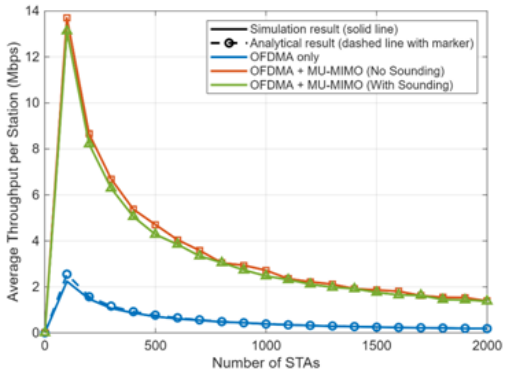


Figure 7.Average per-station throughput :N=8,Nsa=8,Nra=8,lcsi=10

5.5. Average Per-Station Throughput under Non-Saturated Traffic

Figure 7. presents the evaluation of the per-station throughput in a non-saturated environment for various access approaches. The three cases that are considered here include: the use of only OFDMA, the use of MU-MIMO with no channel sounding capability, and the use of MU-MIMO with channel sounding capability. Results show that as the number of STAs increases the

throughput becomes lower. The use of MU-MIMO improves throughput by utilizing spatial multiplexing, whereas channel sounding has limited impact on throughput.

5.6. Impact of Channel Sounding on System Performance

Figure 8. presents various channel sounding frequencies (λcsi = 5, 10, 20, 50) in comparison with the baseline scheme that lacks channel sounding (only OFDMA).

- Throughput: A minor drop can be noted as λcsi increases due to extra costs incurred for collecting CSI. In practice, when using λcsi from 5 to 20, the throughput reduction is small enough.
- Latency: A mild rise in latency can be seen as λcsi rises due to spending a greater share of transmission time on exchanging CSI rather than transmitting the payload.
- Energy Efficiency: Energy efficiency progressively falls under increasingly intensive channel sounding as extra energy is expended in the process of collecting CSI feedback. The value of λcsi = 50 is used here for demonstrating the maximal overhead associated with sounding. It should be noted that it is an impractical sounding rate because it spends almost half of the transmission time on channel sounding, causing a considerable drop in performance metrics. Simulation results align well with the theoretical model, proving its precision.

5.7. Performance of dynamic and static Pareto-based scheduling

Figure 9 depicts the outcome of throughput, delay, and random access probability from simulations and analysis. In static scheduling (Nsa=1,2), the number of resources used remains constant regardless of the buffer state, while the RA-only scheduling works through random access only without scheduling. Dynamic scheduling uses resource allocation depending on the buffer state of every STA and is more efficient since not all allocated resources are active. Split strategy uses static reservation of half of the resources for scheduled access while the other half remains for random access. Specifically, in peak Pareto traffic (N=2000 stations), the proposed dynamic scheduling achieves throughput of 103.97 Mbps, which is 254% higher than the pure RA scheme. Most importantly, even by decreasing the number of random access resources to serve queues, the dynamic mechanism manages to limit the delay of RA stations at 0.0846 s, resulting in 41.7% delay reduction as compared to the strict equal split scheme (0.1453 s).

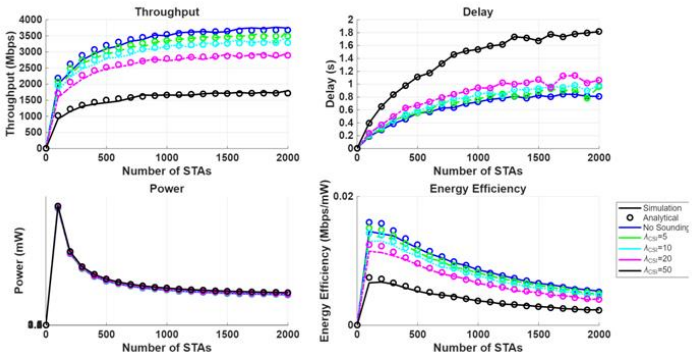


Figure 8.Impact of Channel Sounding Frequency (λcsi) on System Performance:Nsa=10.

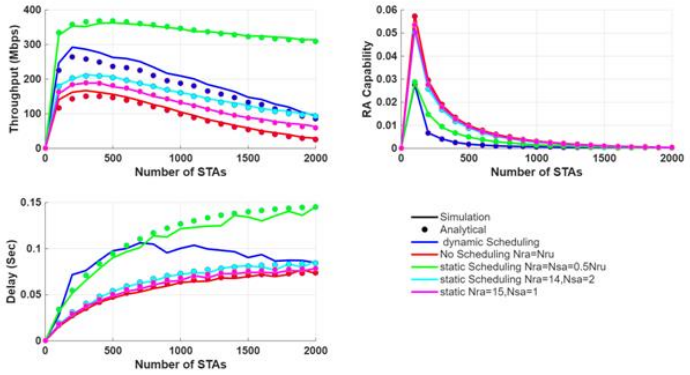


Figure 9.Network performance for λ = 200 and = 4 under different scheduling policies.

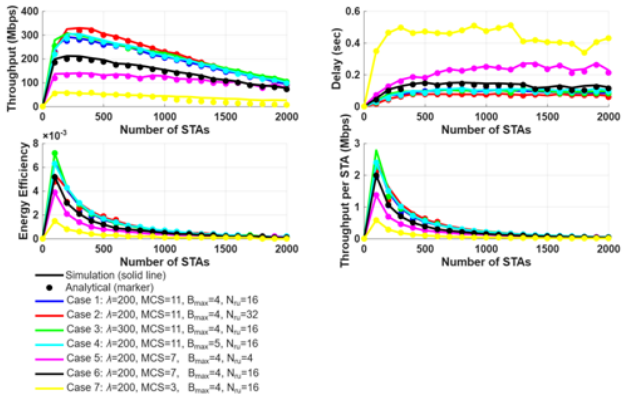


Figure 10.Impact of PHY Parameters on Dynamic Scheduling Performance.

5.8. Impact of Channel Sounding on System Performance

Figure 10 shows the performance of the dynamic scheduling approach on seven test cases, characterized by diverse system parameters. An increase in Nru (Case 2 vs. Case 1) improves throughput and per station throughput without affecting delay. Increased arrival rates (Case 3) positively affect the throughput performance at low to moderate STA numbers but reach saturation for heavy loads, with no impact on the delay performance, illustrating the scheduler robustness. The packet aggregation (Case 4) improves the per station throughput without increasing delay significantly. Quantitatively, during peak network activity, an increase in Nru from 16 (Case 1) to 32 (Case 2) increases simulation throughput from 94.82 Mbps to 99.74 Mbps (an increase of 5.18%), while decreasing delay by 20% (from 0.0846 s to 0.0677 s). Moreover, the importance of the modulation is shown when comparing Cases 5 and 7. Even with 75% less resources available, Case 5 uses a higher MCS number to achieve the peak throughput of 77.12 Mbps, surpassing Case 7 (26.39 Mbps) by 192.2% and decreasing delay by 46.9% (0.2291 s vs. 0.4320 s).

5.9. Comparison between Poisson and Pareto Traffic Models

Figure 11 illustrates the comparison between Poisson and Pareto traffic models with regard to both static and dynamic scheduling mechanisms. There is a good match between simulation and analytical results, confirming the validity of the analytical model. One can notice that dynamic scheduling performs better with higher throughput than static scheduling in all cases due to its

ability to distribute the resources according to the buffer status of active nodes. Additionally, the Pareto traffic model provides higher throughput values in comparison with the Poisson model as its bursty nature enables more efficient exploitation of the transmission opportunities. Quantitatively, during the maximum load period, the developed Pareto-based dynamic scheduling provides peak throughput equal to 50.22 Mbps, outperforming the Pareto-based static scheduling mechanism (28.70 Mbps) by 76%. Under Poisson traffic model, the dynamic scheduling prevents baseline collapse, providing the throughput equal to 9.15 Mbps, surpassing static scheduling mode (2.85 Mbps) by 221%. These direct comparisons confirm mathematically that although dynamic scheduling provides the highest throughput in realistic bursty Pareto load, its ability to optimize the resources leads to important percentage improvements in both traffic models.

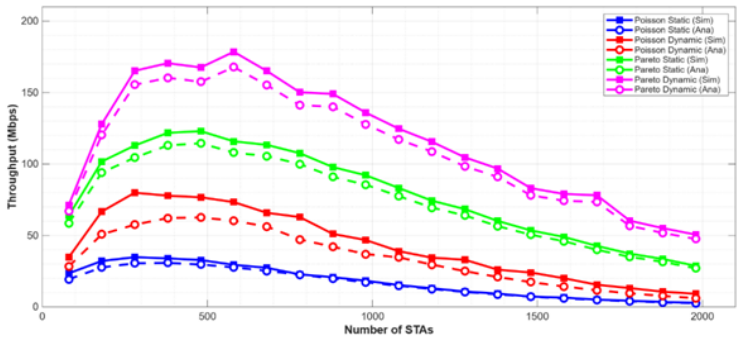


Figure 11.Throughput Comparison under Poisson and Pareto traffic models for static and dynamic Scheduling ($\lambda = 40$, $=5$)

6. CONCLUSION AND FUTURE WORK

This study proposed an integrated model combining theoretical analysis and simulation for assessing the hybrid uplink channel access of IEEE 802.11ax WLANs in non-saturated conditions with Pareto distribution arrival traffic, dynamic RU assignment, MU-MIMO advantage, and channel sounding cost. Our findings revealed that hybrid channel access outperforms in its optimal combination of scheduled and random access channels, with dynamic scheduling contributing greatly to enhanced throughput, access latency, resource efficiency, and energy consumption. The analysis of physical layer parameters, antenna configuration, and sounding interval is valuable for future WLAN design.

Further research could be done in the field of QoS prioritized traffic, adaptive TWT, AI-assisted resource scheduling, and real trace validation with WLANs and Wi-Fi 7 multi-link operation.

CONFLICTS OF INTEREST

The authors declare no conflict of interest.

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